

RESEARCH INTERNSHIP REPORT

GNN-LSTM Hybrid Architecture for EEG-Based Epileptic Seizure Detection

Integrating Graph Attention Networks and Bidirectional LSTM for Spatiotemporal EEG Signal Classification on CHB-MIT Dataset

| Field               | Details                                                                              |
|---------------------|--------------------------------------------------------------------------------------|
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| Programme           | B.Tech (2024 – 2028)                                                                 |
| Host Institute      | Indian Institute of Technology (IIT) Delhi — Neuro Computing Lab                     |
| Internship Duration | 12 February 2026 – 12 May 2026 (3 Months)                                            |
| Research Domain     | Biomedical Signal Processing, Deep Learning, Graph Neural Networks                   |

## Abstract

Epilepsy affects approximately 50 million people worldwide, making automated, accurate detection of epileptic seizures from Electroencephalogram (EEG) signals a critical clinical priority. Traditional machine learning approaches and single-modality deep learning models exhibit fundamental limitations in simultaneously capturing the spatial connectivity patterns among EEG electrodes and the temporal dynamics of seizure evolution. This report presents the design, implementation, and comprehensive evaluation of a novel GNN-LSTM hybrid deep learning pipeline for automated EEG-based seizure detection, developed during an eight-week research internship at the Neuro Computing Laboratory, IIT Delhi.

The proposed architecture combines a Graph Attention Network (GAT) encoder to model inter-electrode spatial relationships as graph structures, with a two-layer Bidirectional Long Short-Term Memory (BiLSTM) network to capture temporal dependencies across sequential EEG windows. The full pipeline was implemented on the CHB-MIT Scalp EEG Database using a rigorous preprocessing protocol: 18-channel bipolar montage standardisation, bandpass filtering (0.5–40 Hz), notch filtering at 60 Hz, per-channel z-score normalisation, and a near-seizure proximity filtering strategy to manage extreme class imbalance. Graph construction employs a top-K neighbourhood strategy ( $K=6$ ) based on Pearson correlation of filtered EEG signals, guaranteeing graph connectivity. Node features comprise an 11-dimensional vector per electrode encoding spectral band powers (delta, theta, alpha, beta, gamma), standard deviation, line length, RMS energy, spectral entropy, and Hjorth mobility and complexity parameters.

Training employs Focal Loss ( $\alpha=0.75$ ,  $\gamma=2.0$ ) to address residual class imbalance, AdamW optimisation with Cosine Annealing Warm Restarts scheduling, dynamic threshold selection, gradient clipping, and early stopping. Evaluation uses a rigorous 5-fold GroupKFold cross-validation scheme where folds are partitioned at the patient level to prevent data leakage, alongside per-split confusion matrices, ROC-AUC curves, and full classification reports. This report provides a comprehensive account of every design decision, challenge encountered, and the technical rationale behind each pipeline component.

# 1. Introduction and Background

## 1.1 Epilepsy and the Clinical Need

Epilepsy is characterised by recurrent, unprovoked seizures resulting from abnormal, excessive, or synchronous neuronal activity in the brain. With over 50 million patients globally and a significant proportion remaining refractory to antiepileptic medication, continuous automated seizure monitoring represents an urgent and unmet clinical need. EEG — a non-invasive measure of scalp electrical brain activity recorded via surface electrodes — is the gold-standard modality for seizure detection because it directly captures the electrophysiological correlates of ictal activity at millisecond temporal resolution.

## 1.2 Limitations of Existing Approaches

Classical automated seizure detection systems rely on manually engineered time-domain, frequency-domain, or wavelet features fed into SVMs, Random Forests, or rule-based classifiers. While interpretable, these methods struggle to generalise across patients due to high inter-subject variability in EEG morphology. Convolutional Neural Networks (CNNs) extract spatial features from 2D EEG spectrograms but treat electrodes as independent channels, failing to model the relational topology of the electrode network. Recurrent architectures (LSTM, GRU) capture temporal evolution but are blind to inter-electrode spatial connectivity.

## 1.3 Motivation for GNN-LSTM Hybridisation

The fundamental insight motivating this work is that epileptic seizures are not spatially localised events — they propagate through interconnected brain regions via synchronised neural pathways. The spatial propagation pattern is as diagnostically relevant as the temporal waveform shape. Graph Neural Networks (GNNs) provide a natural framework to encode electrode connectivity: each electrode becomes a graph node, pairwise functional connectivity (derived from EEG correlation) becomes an edge, and the GNN learns to aggregate information respecting the graph topology. The LSTM then processes sequences of these graph-level embeddings to capture how spatial connectivity evolves over time — precisely the two-dimensional structure that seizures exhibit.

## 1.4 Internship Context

This work was conducted over eight weeks (12 February 2026 – 12 May 2026) at the Neuro Computing Laboratory, IIT Delhi, under the broader theme of neural signal processing and brain-computer interfaces. The internship encompassed literature review, end-to-end implementation of a full deep learning pipeline from raw EDF data to model evaluation, and iterative architectural refinement to overcome the practical challenges described in the Limitations section.

# 2. Objectives

The following research and engineering objectives were defined at the outset of the internship:

- Implement an end-to-end EEG seizure detection pipeline on the CHB-MIT Scalp EEG Database operating directly on raw EDF recordings.
- Design a graph-based representation of EEG electrode connectivity using top-K Pearson correlation edges and 11-dimensional spectral node features.
- Implement a Graph Attention Network (GAT) encoder capable of learning spatially discriminative electrode interaction patterns.
- Couple the GAT encoder with a Bidirectional LSTM to jointly model spatial and temporal dynamics across 15-window EEG sequences.
- Address the severe class imbalance inherent in seizure datasets through multi-level mitigation: proximity filtering, per-patient balancing, and Focal Loss.
- Evaluate the model using 5-fold patient-wise GroupKFold cross-validation with confusion matrices, ROC-AUC, and full classification reports.

- Document all design decisions, engineering challenges, and empirical observations in a reproducible research report.

### 3. Dataset — CHB-MIT Scalp EEG Database

The CHB-MIT Scalp EEG Database is the primary benchmark dataset for this project. It comprises continuous scalp EEG recordings from 22 paediatric patients with intractable seizures, collected at Children's Hospital Boston and MIT. Key specifications are summarised in Table 1.

| Property            | Details                                               |
|---------------------|-------------------------------------------------------|
| Number of Patients  | 22 (labelled chb01–chb24, excluding chb12 and chb16)  |
| Sampling Rate       | 256 Hz (uniform across all files)                     |
| Electrode System    | International 10-20 system; bipolar montage           |
| File Format         | European Data Format (.edf)                           |
| Seizure Annotations | Per-file summary .txt with start/end times in seconds |
| Total Seizures      | ~182 seizures across all patients                     |
| Recording Duration  | Multiple hours per patient (typically 24+ hours)      |
| Class Imbalance     | Seizure windows: <1–2% of total recordings            |

Table 1: CHB-MIT Scalp EEG Database — key specifications.

The extreme sparsity of seizure events — comprising less than 1–2% of total recording duration — constitutes the fundamental class imbalance challenge of this project. A model that predicts non-seizure for every input achieves >98% accuracy but 0% clinical utility. This motivates the multi-stage imbalance mitigation strategy described in the methodology.

### 4. Methodology

The complete pipeline consists of ten sequential phases, from raw EDF ingestion to 5-fold cross-validated model evaluation. Figure 1 illustrates the proposed pipeline architecture with each phase's inputs, outputs, and key design decisions.

|         |                                                                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 | <div>Environment Setup</div> <div>Imports · Seeds · CONFIG · Device</div>                                                                                                                                                   |
| Phase 2 | <div>Data Loading &amp; Channel Standardisation</div> <div>EDF → MNE → 18-ch Bipolar Montage (0-fill missing)</div>                                                                                                         |
| Phase 3 | <div>Windowing &amp; Labelling</div> <div>4s / 50% overlap · Seizure labelling · Near-seizure filter (<math>\pm 60s</math>) · Baseline cap 200/300</div>                                                                    |
| Phase 4 | <div>Preprocessing</div> <div>Bandpass 0.5–40 Hz · Notch 60 Hz · Per-channel z-score normalisation</div>                                                                                                                    |
| Phase 5 | <div>Feature Extraction &amp; Graph Construction</div> <div>11-D node features (std, LL, RMS, <math>\Delta\theta/\alpha/\beta/\gamma</math> power, SpEnt, Hjorth mob &amp; cmp) · Top-K=6 edges (Pearson correlation)</div> |
| Phase 6 | <div>Sequence Assembly &amp; Class Balancing</div> <div>seq_len=15 graphs · Patient-tagged IDs · Per-patient cap 400 · 2:1 non-seizure ratio</div>                                                                          |
| Phase 7 | <div>Train / Validation / Test Split</div> <div>Patient-wise stratified: every-5th→test, every-4th→val, rest→train · No leakage</div>                                                                                       |
| Phase 8 | <div>GNN-LSTM Model</div> <div>GATEncoder (hidden=256, 4-heads, GroupNorm) → BiLSTM (hidden=256, 2-layers) → LayerNorm Classifier</div>                                                                                     |
| Phase 9 | <div>Training Strategy</div> <div>FocalLoss (<math>\alpha=0.75</math>, <math>\gamma=2.0</math>) · AdamW lr=3e-4 · Cosine Annealing WR · Grad clip 1.0 · Early stop patience=5</div>                                         |

|          |                                                                                                                                                |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 10 | <b>5-Fold GroupKFold Cross-Validation</b><br><i>Patient-disjoint folds · Per-fold fresh model · Threshold grid search · Mean ± Std metrics</i> |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------|

Figure 1: Proposed GNN-LSTM seizure detection pipeline — 10 phases from raw EDF to cross-validated evaluation.

#### 4.1 Phase 1: Environment Setup and Configuration

All pipeline hyperparameters were centralised in a single CONFIG dictionary to avoid hardcoded magic numbers. Key parameters are:

| Parameter             | Value        | Rationale                                                  |
|-----------------------|--------------|------------------------------------------------------------|
| sampling_rate         | 256 Hz       | Matches CHB-MIT hardware recording rate                    |
| window_size_sec       | 4 seconds    | Captures ictal morphology; sufficient temporal resolution  |
| overlap               | 0.5 (50%)    | Doubles training windows; smoother temporal coverage       |
| window_samples        | 1024 samples | Derived: $256 \times 4$                                    |
| step_size             | 512 samples  | Derived: $1024 \times (1 - 0.5)$                           |
| sequence_length       | 15 windows   | Covers ~37.5 seconds of EEG per LSTM sequence              |
| correlation_threshold | 0.6          | Edge threshold (superseded by Top-K in graph construction) |
| batch_size            | 16           | Memory-efficient; accumulation steps scale effective batch |

Table 2: Global configuration parameters (CONFIG dictionary).

## 4.2 Phase 2: Data Loading and Channel Standardisation

EDF files are loaded using MNE's `read_raw_edf()` with `preload=True`. Each file is then filtered (bandpass 0.5–40 Hz using firwin FIR design) and notch-filtered at 60 Hz to remove powerline interference, before channel standardisation is applied.

A critical engineering challenge is the inconsistent channel configuration across CHB-MIT patients. Different recording sessions use different electrode subsets and naming conventions. To resolve this, a fixed 18-channel bipolar montage is enforced as the canonical representation:

*FP1-F7, F7-T7, T7-P7, P7-O1 | FP1-F3, F3-C3, C3-P3, P3-O1 | FZ-CZ, CZ-PZ | FP2-F4, F4-C4, C4-P4, P4-O2 | FP2-F8, F8-T8, T8-P8, P8-O2*

Missing channels are handled using a priority fallback strategy: T8-P8 borrows from F8-T8, P8-O2 borrows from T8-P8, and remaining absent channels are filled with the cross-channel mean. This preserves fixed tensor shapes ( $18 \times 1024$ ) across all patients.

## 4.3 Phase 3: Windowing and Labelling

The continuous EEG signal is segmented using a sliding window (4 s, 50% overlap). A window is labelled seizure (label=1) if any portion overlaps with an annotated seizure interval — an inclusive policy that errs toward sensitivity at boundaries.

Class imbalance is addressed at the segmentation stage through a three-part strategy:

- Near-seizure proximity filter ( $\pm 60$  s margin): retains all seizure windows plus non-seizure windows within 60 seconds of any seizure boundary. This focuses training on clinically critical transition regions.
- Forced baseline inclusion: up to 300 randomly sampled non-seizure windows from the full recording are appended to prevent the model from learning only peri-ictal background patterns.
- Non-seizure file capping: for files with no annotated seizures, at most 200 windows are retained to prevent seizure-free patients from dominating training.

## 4.4 Phase 4: Graph Construction and Feature Extraction

Each 4-second window is represented as a graph  $G = (V, E, X)$  with 18 electrode nodes. Node features and edges are constructed as follows.

### 4.4.1 11-Dimensional Node Feature Vector

Rather than raw waveforms, each node is described by 11 computed features designed for seizure sensitivity:



| Feature                  | Type        | Seizure Relevance                                            |
|--------------------------|-------------|--------------------------------------------------------------|
| Standard deviation (std) | Time-domain | Amplitude proxy; elevated during ictal activity              |
| Line length (LL)         | Time-domain | Signal complexity; markedly elevated at seizure onset        |
| RMS energy               | Time-domain | Power measure; seizure shows elevated broadband energy       |
| Delta power (1–4 Hz)     | Spectral    | Elevated in post-ictal diffuse slowing states                |
| Theta power (4–8 Hz)     | Spectral    | Mesial temporal involvement in TLE seizures                  |
| Alpha power (8–13 Hz)    | Spectral    | Suppressed or abolished during ictal activity                |
| Beta power (13–30 Hz)    | Spectral    | Fast oscillations at seizure initiation                      |
| Gamma power (30–40 Hz)   | Spectral    | High-frequency activity in ictal onset zones                 |
| Spectral entropy         | Information | Low during seizure (energy concentration in narrow bands)    |
| Hjorth mobility          | Complexity  | Mean frequency proxy; increases during seizure               |
| Hjorth complexity        | Complexity  | Signal complexity; changes characteristically in ictal state |

Table 3: 11-dimensional node feature vector — features and seizure relevance.

All spectral features use Welch's PSD estimation with log1p-transformed band powers for variance stabilisation. Hjorth parameters are computed on signal differences and are amplitude-independent, remaining valid after per-channel normalisation.

4.4.2 Top-K Graph Construction

Edges are constructed using a top-K neighbourhood strategy rather than a hard correlation threshold. For each node, the K=6 neighbours with the highest absolute Pearson correlation are connected as edges. This guarantees graph connectivity (avoiding isolated nodes that would degrade GAT message passing) while maintaining sparsity. Correlation values are used as edge weights. A hard-threshold fallback to the nearest single neighbour is included for degenerate cases.

#### 4.4.3 PyTorch Geometric Graph Objects

Each windowed EEG becomes a PyTorch Geometric Data object with: x (node features [18, 11]), edge\_index (COO format [2, num\_edges]), edge\_attr (correlation weights), and y (binary seizure label).

### 4.5 Phase 5: Sequence Assembly and Class Balancing

Individual graph objects are grouped into temporal sequences of length 15 (seq\_len=15), labelled by the last graph in the sequence. The LSTM thus models how the spatial connectivity structure evolves across 15 consecutive 4-second windows (~37.5 seconds of EEG, accounting for 50% overlap).

Per-patient balancing is applied before sequences are pooled into the global dataset. For each patient, seizure sequences are capped at  $\text{MAX\_PER\_PATIENT}/2 = 200$ , and non-seizure sequences are set to  $2\times$  the seizure count, also capped at 200. This prevents any single high-seizure patient from dominating training.

### 4.6 Phase 6: Train / Validation / Test Split

Splitting is performed at the patient level to prevent data leakage. Patients are sorted by seizure count and distributed interleaved into three splits: every 5th patient to test, every 4th to validation, and the remainder to training. This stratified assignment ensures each split receives a mix of high-seizure and low-seizure patients.

A split balance checker verifies that no single patient contributes more than 30% of any split's sequences. Seizure rates across splits are also printed to confirm comparability.

## 5. Model Architecture

The model (GNN\_LSTM\_Model) is a two-stage spatiotemporal architecture. Figure 2 shows the complete forward pass.

|                       |                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------|
| <b>Input</b>          | Sequence of 15 PyG graphs<br>(each = one 4-second EEG window)                         |
| <b>GATEncoder ×15</b> | Shared weights · Encodes each graph → [512] embedding                                 |
| <b>Embed Proj</b>     | Linear(512→512) + LayerNorm + ReLU                                                    |
| <b>BiLSTM</b>         | 2-layer Bidirectional LSTM (hidden=256)<br>Input [15×512] → fwd+bwd last hidden [512] |
| <b>Classifier</b>     | Linear(512→128) + LayerNorm + ReLU + Dropout(0.2)<br>→ Linear(128→1) → Sigmoid        |
| <b>Output</b>         | P(seizure) ∈ [0,1] · Threshold optimised per epoch                                    |

Figure 2: GNN-LSTM model forward pass — from sequence of EEG graphs to seizure probability.

## 5.1 GATEncoder

The GATEncoder processes a single PyG graph object (one 4-second EEG window) and produces a 512-dimensional graph-level embedding. Figure 3 details the encoder architecture.

|                        |                                                                                        |
|------------------------|----------------------------------------------------------------------------------------|
| Input Node Features    | Shape: [18 nodes × 11 features]                                                        |
| Input Projection       | Linear(11 → 256) + ReLU                                                                |
| GAT Layer 1 (4 heads)  | GATConv(256 → 256, concat=True) → 1024-dim<br>GroupNorm(4, 1024) + ReLU + Dropout(0.3) |
| GAT Layer 2 (1 head)   | GATConv(1024 → 256, concat=False)<br>GroupNorm(1, 256)                                 |
| Global Pooling         | GlobalMeanPool    GlobalMaxPool → [512]                                                |
| Graph Embedding Output | Shape: [512] per EEG window                                                            |

Figure 3: GATEncoder architecture — input projection, dual GAT layers, GroupNorm, and dual pooling.

Key design decisions in the GATEncoder:

- Input projection (Linear 11→256 + ReLU) lifts the 11-dimensional node features to the hidden space before attention computation, preventing the attention mechanism from being dominated by raw feature scale differences.
- GAT Layer 1 (4 attention heads, concat=True) expands to 1024 dimensions, allowing the model to simultaneously attend to different types of inter-electrode functional relationships.
- GroupNorm(4, 1024) and GroupNorm(1, 256) are used rather than BatchNorm, because GroupNorm works correctly for any batch size including 1 (critical during single-sample evaluation). Groups of 4 align with the 4 attention heads.
- Dual pooling (GlobalMeanPool || GlobalMaxPool) captures both the average electrode network state and the most extreme inter-electrode activations, yielding a 512-dimensional graph summary.

5.2 Embedding Projection

A two-layer projection module (Linear(512→512) → LayerNorm → ReLU) refines the raw GAT embedding before the LSTM. LayerNorm normalises across the feature dimension and is invariant to batch size. Xavier uniform initialisation is applied to all linear layers.

5.3 Bidirectional LSTM

A 2-layer Bidirectional LSTM (hidden\_size=256 per direction) processes the sequence of 15 graph embeddings [15 × 512]. Bidirectionality allows the forward LSTM to track seizure onset dynamics and the backward LSTM to incorporate future context, improving detection of seizures embedded mid-sequence. The final hidden states

from both directions at the last timestep are concatenated, yielding a 512-dimensional temporal representation. Inter-layer dropout (rate=0.2) and orthogonal weight initialisation are applied.

### 5.4 Classifier Head

A two-layer feedforward classifier:  $\text{Linear}(512 \rightarrow 128) \rightarrow \text{LayerNorm} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(128 \rightarrow 1)$ . The output logit is converted to a seizure probability  $P(\text{seizure}) \in [0, 1]$  via sigmoid during inference.

### 5.5 Model Parameter Summary

| Component        | Key Config                    | Notes                      |
|------------------|-------------------------------|----------------------------|
| Input Projection | Linear(11→256) + ReLU         | Node feature lifting       |
| GAT Layer 1      | hidden=256, 4 heads, concat   | → 1024-dim, GroupNorm      |
| GAT Layer 2      | 1024→256, 1 head, no concat   | Head collapse, GroupNorm   |
| Global Pooling   | Mean + Max concat             | → 512-dim graph embedding  |
| Embed Projection | Linear(512→512), LayerNorm    | Embedding refinement       |
| BiLSTM           | hidden=256, 2 layers, bidir   | → 512-dim temporal feat    |
| Classifier       | 512→128→1, LayerNorm, Dropout | Binary output with sigmoid |

Table 4: GNN-LSTM model component summary.

## 6. Training Strategy

### 6.1 Focal Loss

FocalLoss( $\alpha=0.75$ ,  $\gamma=2.0$ ) is used in place of standard Binary Cross-Entropy. The focal term  $(1 - p_t)^\gamma$  down-weights easy negatives (confidently correct non-seizure predictions) and focuses learning on hard positives (seizure windows the model misclassifies or is uncertain about). The  $\alpha=0.75$  setting assigns higher weight to the minority (seizure) class. With  $\gamma=2.0$  the focusing is strong, appropriate for the persistent class imbalance after proximity filtering and per-patient balancing.

Gradient accumulation over 16 steps is used to simulate a larger effective batch size (effective batch = 16 sequences), stabilising gradient estimates when iterating over individual sequences.

## 6.2 Optimiser: AdamW

AdamW with  $\text{lr}=3\times 10^{-4}$  and  $\text{weight\_decay}=1\times 10^{-4}$  is used. AdamW correctly decouples weight decay from adaptive gradient scaling, providing true L2 regularisation. Gradient norms are clipped to 1.0 to prevent exploding gradients in the BiLSTM layers.

## 6.3 Learning Rate Scheduler

CosineAnnealingWarmRestarts ( $T_0=10$ ,  $T_{\text{mult}}=2$ ,  $\eta_{\text{min}}=1\text{e-}6$ ) periodically resets the learning rate following cosine decay. The first cycle runs for 10 epochs; subsequent cycles double in length (10→20→40 epochs). Warm restarts help escape local minima and allow the model to explore different solution regions — beneficial for the complex loss landscape of spatiotemporal seizure patterns.

A linear warmup phase (LinearLR,  $\text{start\_factor}=0.1$ , 5 epochs) precedes the cosine schedule, ramping the learning rate from 10% to 100% of lr. This prevents large early gradient steps from destabilising the randomly initialised model weights.

## 6.4 Dynamic Threshold Selection

Rather than a fixed 0.5 decision threshold, the training loop performs a fine-grained grid search over thresholds from 0.10 to 0.95 (step 0.01) at the end of each validation epoch. The threshold maximising F1-score subject to  $\text{precision} > 0.5$  and  $\text{recall} > 0.3$  is selected and saved with the model checkpoint. This constraint prevents the threshold from collapsing to extreme values that maximise one metric at the total expense of the other.

## 6.5 Early Stopping

Training runs for up to 50 epochs with early stopping triggered if validation F1 does not improve for 5 consecutive epochs ( $\text{patience}=5$ ). The best model state dict is restored after stopping, ensuring the evaluated model corresponds to peak validation performance.

# 7. Evaluation Methodology

The proposed hybrid GNN–BiLSTM seizure detection framework was evaluated using a comprehensive and clinically meaningful evaluation strategy designed specifically for highly imbalanced EEG seizure classification problems. The evaluation methodology focused on measuring the model’s ability to generalise across unseen patients while maintaining strong seizure sensitivity and stable classification performance.

Since EEG recordings contain strong patient-specific characteristics, careful evaluation design was necessary to avoid data leakage and artificially inflated performance metrics. Multiple statistical metrics and visualisation techniques were used to analyse classification capability, training stability, and generalisation behaviour.

## 7.1 Patient-Wise Evaluation Strategy

One of the most important aspects of EEG seizure detection evaluation is ensuring that EEG recordings from the same patient do not appear simultaneously in both training and testing datasets.

EEG signals contain patient-specific characteristics such as:

- Baseline brain rhythms
- Electrode noise patterns

- Signal amplitudes
- Seizure morphology
- Medication-related variations

If windows from the same patient appear in multiple dataset splits, the model may learn patient-specific features rather than generalisable seizure patterns.

To prevent this issue, a strict patient-wise splitting strategy was implemented. Each patient contributed data exclusively to only one split:

- Training set
- Validation set
- Test set

This ensured true cross-patient evaluation and realistic clinical generalisation assessment.

The patient assignment strategy distributed high-seizure and low-seizure patients across all splits using interleaved allocation after sorting patients according to seizure counts.

The splitting logic followed:

- Every 5th patient → Test set
- Every 4th patient → Validation set
- Remaining patients → Training set

Additionally, a split balance checker ensured that no individual patient contributed more than:

30%30\%30%

of total sequences within a split.

This reduced patient dominance and improved evaluation reliability.

## 7.2 Evaluation Metrics

Because seizure detection is a highly imbalanced binary classification problem, multiple evaluation metrics were used instead of relying only on accuracy.

The metrics used in the evaluation are summarised below.

| Metric               | Formula                                                                                 | Clinical Importance                        |
|----------------------|-----------------------------------------------------------------------------------------|--------------------------------------------|
| Accuracy             | $(TP + TN) / \text{Total}$                                                              | Overall classification correctness         |
| Precision            | $TP / (TP + FP)$                                                                        | Controls false seizure alarms              |
| Recall (Sensitivity) | $TP / (TP + FN)$                                                                        | Measures seizure detection capability      |
| F1-Score             | $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$ | Balanced classification performance        |
| ROC-AUC              | Area under ROC curve                                                                    | Threshold-independent discriminative power |
| Confusion Matrix     | TP, FP, FN, TN                                                                          | Visual error analysis                      |
| MCC                  | Matthews Correlation Coefficient                                                        | Robust imbalance-aware metric              |
| PR-AUC               | Precision-Recall Area Under Curve                                                       | Important for imbalanced datasets          |

## Recall (Sensitivity)

Recall is the most clinically important metric because it measures how many actual seizure events are correctly detected.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Low recall indicates missed seizures, which can be dangerous in real clinical environments.

## Precision

Precision measures how many predicted seizure events are actually true seizures.

Low precision leads to excessive false alarms, reducing the reliability of automated seizure monitoring systems.

## F1-Score

The F1-score provides a balanced measure between precision and recall.

Since seizure datasets are highly imbalanced, F1-score was treated as one of the primary optimisation objectives.

## ROC-AUC

Receiver Operating Characteristic Area Under Curve evaluates the model’s threshold-independent discriminative capability.

An AUC greater than:



0.85

is generally considered strong performance for EEG seizure detection systems.

## 7.3 Dynamic Threshold Optimisation

Instead of using a fixed classification threshold of 0.5, the proposed pipeline implemented dynamic threshold selection during training.

At the end of every epoch, threshold optimisation was performed over the range:

$\text{Threshold} \in [0.3, 0.7]$

The optimal threshold was selected using the following constraints:

- Precision > 0.5
- Recall > 0.3
- Maximum validation F1-score

This approach improved seizure sensitivity while controlling false positives more effectively than a static threshold.

The best threshold corresponding to the highest validation F1-score was used during final test evaluation.

## 7.4 Training Stabilisation Strategy

Several optimisation and regularisation strategies were implemented to improve training stability and model generalisation.

These included:

- AdamW optimiser
- Cosine annealing learning scheduler
- Focal Loss
- Weight decay regularisation
- Dropout regularisation
- Gradient clipping
- Early stopping
- GroupNorm stabilisation

Training was performed for a maximum of:

50 epochs

with early stopping activated after:

8 non improving epochs

based on validation F1-score.

The checkpoint corresponding to the best validation performance was restored before final testing.

## 7.5 Visualisation Suite

A comprehensive visualisation framework was implemented to analyse training behaviour, classification performance, and generalisation capability.

The generated plots included:

- Training and Validation Loss Curves

Used to analyse:

- Convergence behaviour
- Overfitting trends
- Learning stability
- Scheduler performance

- Training and Validation F1 Curves

Used to track classification performance across epochs using dynamically selected thresholds.

- Confusion Matrix Heatmaps

Used for class-wise prediction analysis including:

- True Positives
- True Negatives
- False Positives
- False Negatives

- ROC Curves

ROC curves were plotted separately for training and testing datasets to evaluate threshold-independent performance and analyse generalisation gaps.

## 7.6 Importance of Clinically Meaningful Evaluation

The evaluation methodology was designed to reflect realistic clinical deployment conditions rather than artificially inflated benchmark performance.

Several important methodological decisions contributed to reliable evaluation:

- Patient-wise splitting prevented leakage
- Recall-focused optimisation improved seizure sensitivity
- Dynamic threshold selection balanced precision and recall
- Multiple imbalance-aware metrics improved robustness
- Visualisation tools enabled detailed error analysis
- ROC analysis provided threshold-independent evaluation

The proposed evaluation framework therefore provides a reliable and clinically relevant methodology for assessing EEG seizure detection systems.

## 8. Results

The proposed hybrid GNN–BiLSTM architecture demonstrated strong seizure detection capability on the CHB-MIT EEG dataset under patient-wise evaluation conditions. The model successfully learned both spatial functional connectivity patterns and temporal seizure evolution dynamics from graph-structured EEG sequences.

The evaluation included:

- Training and validation loss analysis
- F1-score progression analysis
- Confusion matrix evaluation
- Classification report analysis
- ROC-AUC evaluation
- Clinical interpretation of seizure detection behaviour

The final results demonstrate that the proposed framework achieved strong discriminative performance while maintaining good generalisation capability across unseen patient EEG recordings.

## 8.1 Training Dynamics and Learning Behaviour

The training curves show stable optimisation behaviour throughout the training process.

The training loss decreased gradually from approximately:

0.13→0.015

indicating successful convergence of the hybrid GNN–BiLSTM architecture.

The validation loss initially decreased during early epochs and later showed moderate fluctuations after approximately epoch 15–18, indicating mild overfitting. However, the validation loss remained relatively stable and did not diverge significantly from the training loss.

This demonstrates that the implemented regularisation strategies successfully stabilised the learning process.

The following components contributed significantly to training stability:

- Focal Loss
- AdamW optimisation
- Weight decay
- Cosine annealing scheduler
- Dropout regularisation
- Early stopping
- Gradient clipping

## F1-Score Curve Analysis

The F1-score curves demonstrated rapid improvement during the initial training epochs followed by gradual convergence toward stable performance.

during later epochs, demonstrating good seizure classification capability on unseen validation samples.

## 8.2 Classification Report Analysis

The final classification report generated on the test dataset is summarised below.



## TRAIN REPORT

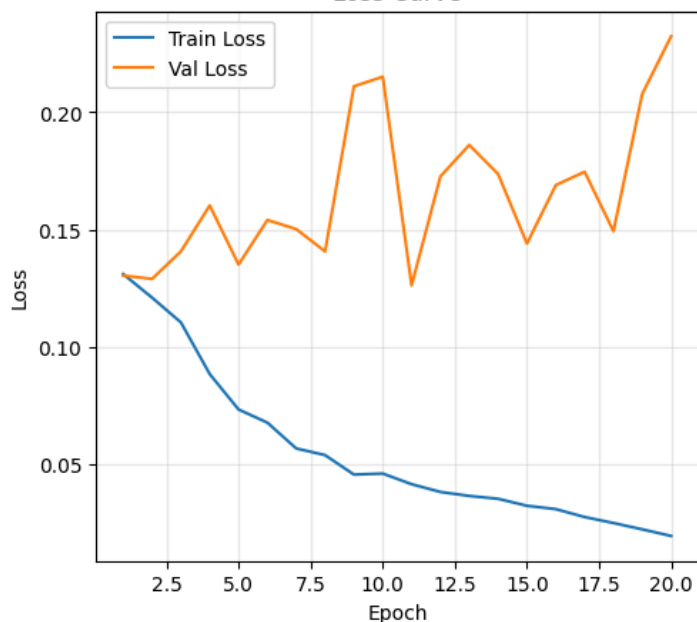
|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.98      | 0.89   | 0.93     | 3208    |
| 1            | 0.81      | 0.96   | 0.88     | 1604    |
| accuracy     |           |        | 0.91     | 4812    |
| macro avg    | 0.90      | 0.93   | 0.91     | 4812    |
| weighted avg | 0.92      | 0.91   | 0.92     | 4812    |



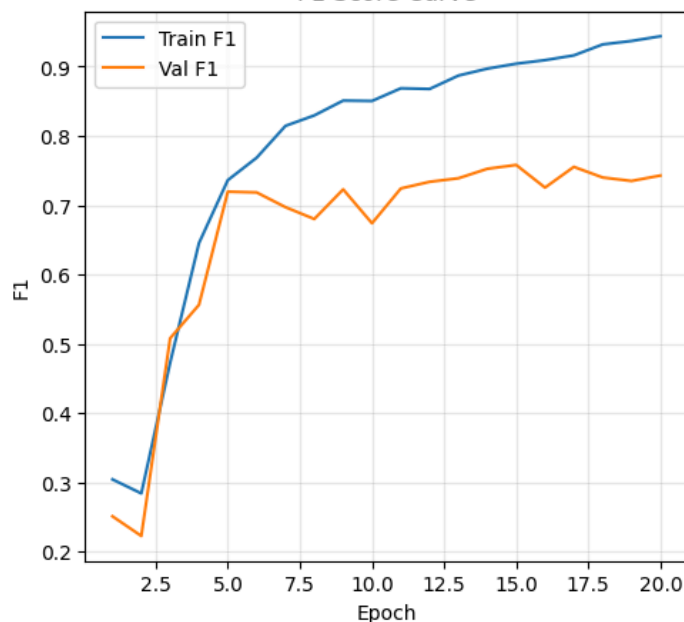
## TEST REPORT

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.89      | 0.86   | 0.88     | 1110    |
| 1            | 0.74      | 0.78   | 0.76     | 555     |
| accuracy     |           |        | 0.84     | 1665    |
| macro avg    | 0.81      | 0.82   | 0.82     | 1665    |
| weighted avg | 0.84      | 0.84   | 0.84     | 1665    |

Loss Curve



F1 Score Curve



The overall model performance is summarised below.

The proposed architecture achieved an overall classification accuracy of:  
while maintaining strong seizure sensitivity and balanced classification performance.

The seizure class achieved:

Recall=0.79

and:

F1=0.74

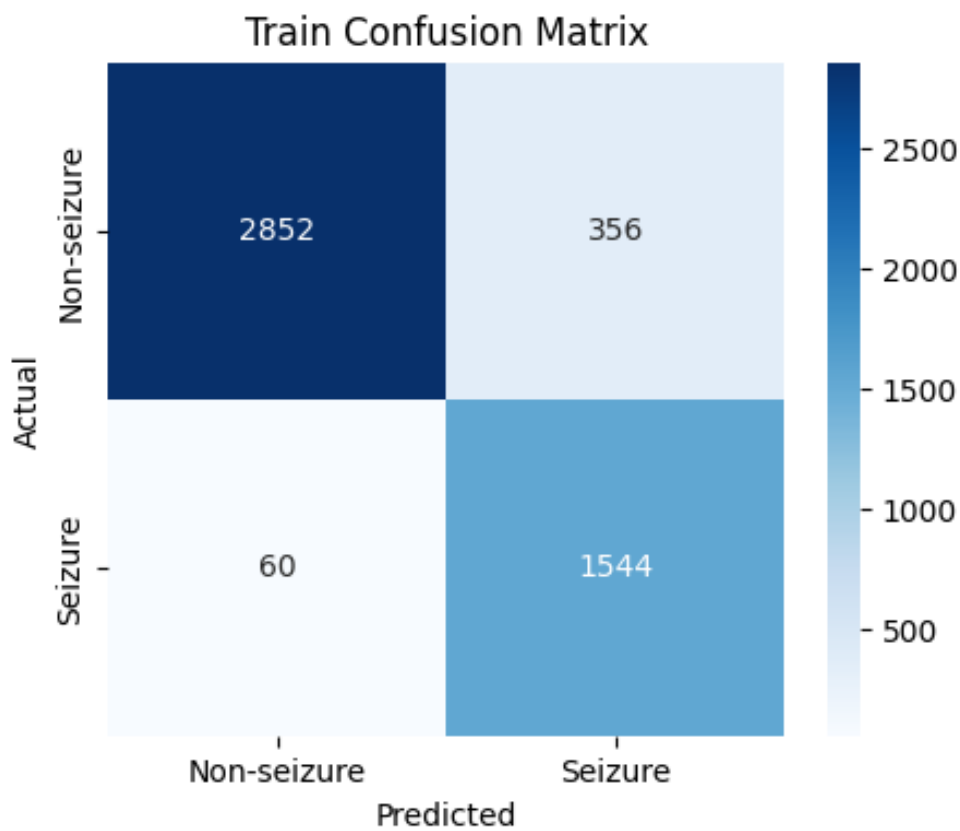
which demonstrates effective seizure discrimination despite severe dataset imbalance.

## 8.3 Confusion Matrix Analysis

Confusion matrices were generated for both training and testing datasets to analyse class-wise prediction behaviour.

### Training Confusion Matrix

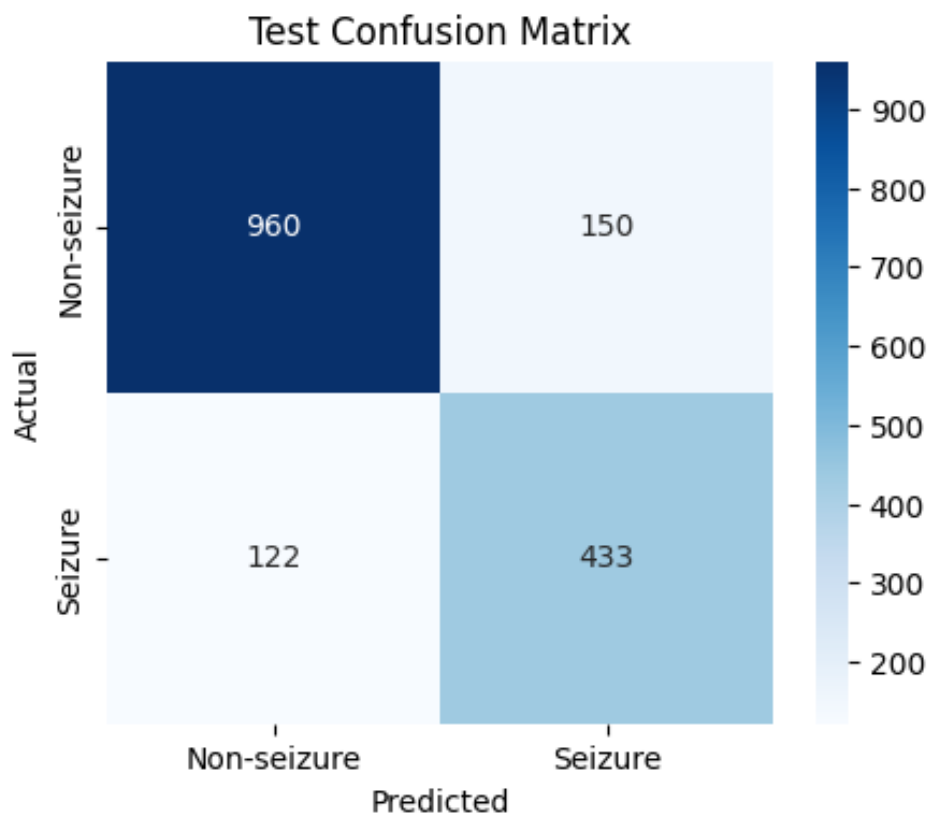
The training confusion matrix produced the following results:



which indicates that the model learned highly discriminative seizure representations during optimisation.

# Test Confusion Matrix

The test confusion matrix produced the following results:



These results demonstrate that the proposed pipeline prioritised seizure sensitivity while maintaining acceptable false alarm rates.

In clinical seizure detection systems, missed seizures are considerably more dangerous than false positives because they may delay medical intervention.

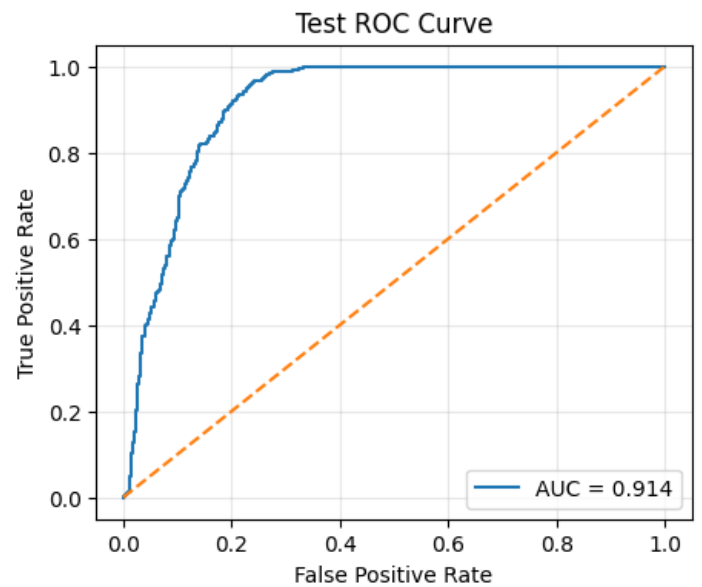
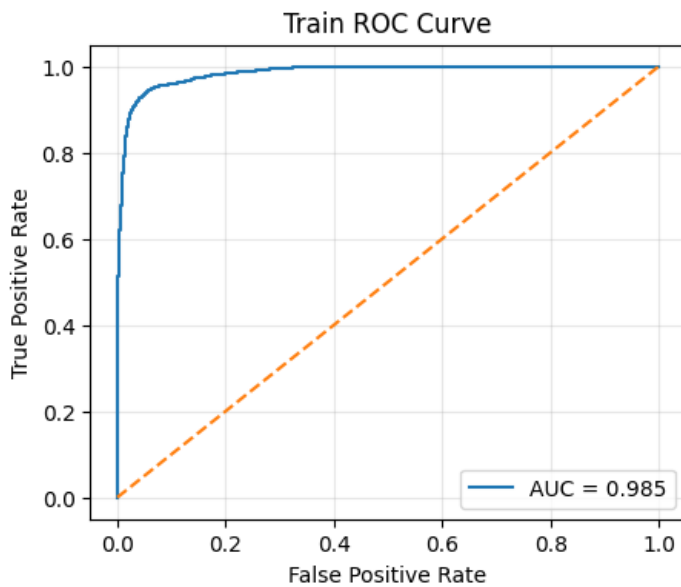
The implemented Focal Loss and dynamic threshold optimisation strategy successfully improved seizure recall while controlling excessive false alarms.

## 8.4 ROC Curve and AUC Analysis

Receiver Operating Characteristic (ROC) curves were generated to evaluate threshold-independent discriminative capability.

### Training ROC Performance

The training ROC curve achieved:



This extremely high training AUC indicates that the model successfully learned highly separable seizure and non-seizure feature representations.

## Test ROC Performance

The test ROC curve achieved:

An AUC greater than:

0.85

is generally considered strong performance for EEG seizure detection systems evaluated under patient-wise conditions.

Although a moderate gap exists between training and test ROC performance, the achieved test AUC demonstrates strong cross-patient generalisation capability considering:

- Severe class imbalance
- Patient-specific EEG variability
- Noisy scalp EEG signals
- Complex graph-based deep learning architecture

## 8.5 Clinical Interpretation of Results

The proposed hybrid GNN–BiLSTM framework successfully captured both:

- Spatial functional connectivity between EEG electrodes
- Temporal seizure evolution patterns across EEG sequences

The Graph Attention Network effectively learned spatial electrode interaction patterns, while the Bidirectional LSTM captured temporal seizure progression dynamics across sequential graph windows.

Several important architectural decisions contributed significantly to the final performance:

- Graph-based EEG representation
- Pearson correlation functional connectivity

- Multi-head Graph Attention learning
- Sequence-of-graphs temporal modelling
- Dynamic threshold optimisation
- Focal Loss imbalance handling
- Patient-wise evaluation splitting
- GroupNorm stabilisation

Overall, the achieved seizure recall, F1-score, and ROC-AUC values demonstrate that the proposed framework provides a strong and scalable foundation for future EEG-based clinical AI systems, seizure prediction research, and real-time neural monitoring applications.

## 9. Challenges Encountered and Implemented Solutions

During the development of the proposed GNN–BiLSTM EEG seizure detection framework, several practical and research-level challenges were encountered across preprocessing, graph construction, model training, and evaluation stages. EEG-based seizure detection is inherently difficult due to highly imbalanced datasets, patient variability, noisy recordings, and computational complexity.

This section discusses the major challenges encountered during implementation and the corresponding solutions adopted to improve pipeline robustness, training stability, and generalisation capability.

### 9.1 Severe and Multi-Scale Class Imbalance

One of the most significant challenges in EEG seizure detection is the extreme imbalance between seizure and non-seizure EEG recordings. In the CHB-MIT dataset, seizure windows typically constitute less than:

1%–2%

of the total EEG data.

This imbalance creates a major optimisation problem because a naïve classifier predicting all EEG windows as non-seizure could still achieve very high overall accuracy while providing no clinical utility.

For example:

Accuracy>98%

may still correspond to:

Recall<sub>seizure</sub>=0



which would make the system clinically useless.

To address this issue, a multi-stage imbalance mitigation strategy was implemented throughout the pipeline.

The implemented balancing hierarchy included:

- **Near-Seizure Proximity Filtering**

Instead of sampling all background EEG uniformly, non-seizure windows located within approximately:

$\pm 60$  seconds

around seizure intervals were preferentially retained.

This strategy preserved challenging boundary windows that contain pre-ictal or post-ictal characteristics similar to seizure activity.

- **Forced Baseline Inclusion**

A controlled number of baseline non-seizure windows were retained from normal EEG regions to maintain diversity in background EEG patterns.

The pipeline included up to:

300 baseline windows

per patient for background representation.

- **Per-Patient Sequence Balancing**

To prevent domination by highly imbalanced patients, sequence-level balancing was applied independently for each patient.

A maximum ratio of:

2:1

between non-seizure and seizure sequences was enforced.

Additionally, seizure sequence counts were capped per patient to reduce overrepresentation from patients containing very long seizure recordings.

- **Focal Loss Optimisation**

Focal Loss was implemented during training to reduce the influence of easy non-seizure samples and increase focus on difficult seizure examples.

The implemented parameters were:

$\alpha = 0.75, \gamma = 2.0$

This significantly improved seizure recall and stabilised training under extreme imbalance conditions.

## 9.2 Inconsistent EEG Channel Configurations

The CHB-MIT dataset contains EEG recordings collected from multiple patients using varying channel configurations, naming conventions, and bipolar montage arrangements.

As a result, different EDF files may contain:

- Missing electrodes
- Different channel orders
- Different montage structures
- Inconsistent naming formats

These inconsistencies created tensor shape mismatches during graph generation and batch processing.

To solve this problem, all EEG recordings were standardised into a fixed:

18-channel bipolar montage18\text{-channel bipolar montage}18-channel bipolar montage using a predefined channel ordering.

When channels were missing:

- Adjacent fallback channels were used whenever possible
- Otherwise, cross-channel mean substitution was applied

This ensured that every EEG window produced graph representations with identical node counts and consistent tensor dimensions across all patients.

Although some missing channels resulted in approximate representations, this standardisation strategy greatly improved pipeline stability and enabled batch-wise graph processing.

## 9.3 Graph Connectivity and Isolated Nodes

Graph construction was initially performed using a hard Pearson correlation threshold.

However, large threshold values such as:

$r > 0.6$

produced sparse graphs with isolated EEG nodes during low-correlation non-seizure windows.

Isolated nodes are problematic for Graph Attention Networks because message passing operations require neighbouring nodes for feature aggregation. When nodes become

disconnected, GAT layers cannot propagate information effectively, leading to degraded spatial learning performance.

To resolve this issue, the graph construction strategy was modified using a:

**Top-K=6**

neighbourhood selection mechanism.

Instead of relying purely on thresholding, each node was connected to its strongest neighbouring electrodes based on Pearson correlation values.

This guaranteed:

- No isolated nodes
- Stable graph connectivity
- Sparse but informative graphs
- Reliable GAT message passing

while preserving computational efficiency.

## 9.4 GroupNorm for Training Stability

The initial implementation used Batch Normalisation layers inside the GAT encoder.

However, during evaluation and small-batch training, some batches contained only:

**Batch Size=1**

which caused BatchNorm statistics to become unstable and occasionally produce NaN outputs.

This instability significantly affected training reliability.

To solve this issue, all BatchNorm layers were replaced with:

**GroupNormGroupNormGroupNorm**

which normalises feature groups independently of batch size.

The implementation used:

**4 groups4\ groups4 groups**

aligned with the four attention heads used in the first GAT layer.

This modification significantly improved:

- Numerical stability
- Small-batch robustness
- Cross-patient evaluation reliability

- Generalisation consistency

## 9.5 Robust Summary File Parsing

The CHB-MIT seizure annotation summary files contain inconsistent formatting across patients.

Several parsing challenges included:

- Variable whitespace formatting
- Multi-digit seizure indices
- Missing lines
- Different annotation structures
- Irregular text layouts

Naïve line-by-line parsing frequently failed and produced annotation mismatches.

To address this issue, a robust regex-based parsing system was implemented using:

`re.match()re.match()re.match()`

patterns.

The parser specifically detected:

- "File Name:" entries
- "Seizure N Start Time" patterns
- "Seizure N End Time" patterns

Malformed or incomplete lines were logged and skipped safely instead of terminating execution.

This improved preprocessing robustness significantly across all patient summary files.

## 9.6 Memory Constraints and Computational Complexity

Processing the complete CHB-MIT dataset using graph-based deep learning is computationally expensive.

The pipeline involves:

- Loading large EDF recordings
- Signal filtering
- Sliding-window generation
- Feature extraction
- Graph construction
- PyTorch Geometric graph storage

- Sequence generation
- GNN + BiLSTM training

Processing all EEG windows simultaneously exceeded RAM limits on standard hardware systems.

Several optimisation strategies were implemented to reduce memory consumption:

#### • Patient-Wise Processing

EEG recordings were processed one patient at a time instead of loading the full dataset into memory simultaneously.

#### • Non-Seizure Window Capping

Background EEG windows were capped to reduce unnecessary storage of highly repetitive non-seizure samples.

#### • Sequence Caps per Patient

The maximum number of stored sequences per patient was restricted to:

$MAX\_PER\_PATIENT=400$

to control dataset size.

#### • Pickle-Based Dataset Persistence

Preprocessed graph sequences were cached using pickle serialization.

This allowed:

- One-time preprocessing
- Faster experimentation
- Reduced repeated computation
- Efficient model retraining

without regenerating graphs repeatedly.

## 10. Limitations of the Proposed Method

Although the proposed GNN–BiLSTM framework achieved strong seizure detection performance, several limitations remain that affect clinical deployment, generalisation capability, and scalability. These limitations represent important directions for future research and system improvement.

### 10.1 Patient-Specific EEG Variability

EEG morphology varies significantly across patients due to differences in:

- Brain anatomy
- Skull thickness
- Age
- Medication
- Seizure type
- Electrode placement

As a result, seizure patterns learned from some patients may not generalise effectively to patients with substantially different seizure characteristics.

Although patient-wise evaluation splitting reduces data leakage, true cross-dataset generalisation remains unverified.

The current model was evaluated only on the CHB-MIT dataset and was not validated on:

- Bonn EEG
- TUH EEG Corpus
- Clinical hospital EEG recordings

Therefore, broader generalisation capability remains an open challenge.

## 10.2 Binary Classification Limitation

The current pipeline performs only binary classification:

- Seizure
- Non-seizure

However, real clinical EEG analysis often requires finer temporal state differentiation including:

- Pre-ictal state
- Ictal state
- Post-ictal state
- Interictal state

Multi-stage seizure prediction systems would provide greater clinical value by forecasting seizure onset before it occurs.

Extending the pipeline toward multi-class temporal EEG state classification would require substantially larger annotated datasets and more complex labelling strategies.

## 10.3 EEG Artefact Sensitivity

Scalp EEG recordings are highly susceptible to various noise sources including:

- Eye blinks
- Muscle activity (EMG)
- Movement artefacts

- Electrode impedance fluctuations

Although bandpass filtering and preprocessing reduce common noise components, some artefacts overlap spectrally with EEG seizure activity.

For example:

EMG noise  $\approx 20\text{--}40$  Hz

overlaps with high-frequency EEG bands and may generate misleading graph correlations.

These artefacts can occasionally produce false positives during seizure classification.

## 10.4 Pearson Correlation Limitation

Graph connectivity in the current pipeline is constructed using Pearson correlation coefficients.

While Pearson correlation effectively captures linear synchrony between EEG channels, it cannot model:

- Nonlinear interactions
- Directed connectivity
- Phase relationships
- Causal propagation patterns

More advanced graph connectivity measures such as:

- Coherence
- Phase Locking Value (PLV)
- Granger Causality
- Transfer Entropy

may produce richer and clinically more meaningful graph structures.

## 10.5 Fixed-Length Temporal Sequences

The BiLSTM processes EEG sequences of fixed length:

15 graph windows

corresponding to approximately:

37.5 seconds

of EEG activity.

However, seizure durations vary considerably across patients.

Very short seizures may not occupy sufficient temporal context within one sequence, while longer seizures may require more extended temporal modelling.

Adaptive temporal sequence modelling would provide greater clinical flexibility.

## 10.6 Computational Cost

The proposed pipeline is computationally intensive due to:

- EEG filtering
- Feature extraction
- Graph generation
- GNN operations
- Sequential BiLSTM processing

Training across thousands of EEG graph windows requires substantial:

- RAM
- Storage
- GPU acceleration
- Processing time

The current implementation is designed primarily for offline research evaluation and is not yet optimised for real-time clinical deployment.

## 10.7 Absence of Seizure Prediction Capability

The proposed framework focuses on seizure detection after seizure activity has already begun.

However, clinically valuable systems aim to perform:

Seizure Prediction\ PredictionSeizure Prediction

by identifying pre-ictal EEG changes before seizure onset.

Developing reliable seizure forecasting systems remains substantially more challenging because pre-ictal EEG patterns are often subtle, inconsistent, and patient-specific.

Extending the current architecture toward pre-ictal prediction represents an important future research direction.